

Wednesday Warm Up: DC circuits, and Kirchhoff's Rules

Prof. Jordan C. Hanson

February 18, 2026

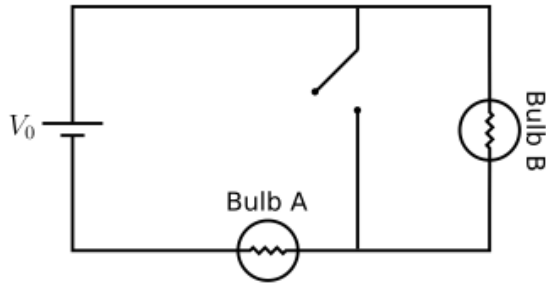


Figure 1: Two light bulbs connected to an EMF V_0 .

1 Memory Bank

- $V = IR$... Ohm's Law.
- $P = IV = I^2R = V^2/R$... Power consumption.
- Kirchhoff's Rule #1: Total current into a junction equals total current out of a junction.
- Kirchhoff's Rule #2: The total change in voltage around a loop is zero.

2 Resistors in Series and Parallel, Kirchhoff's Rules

1. Consider Fig. 1. (a) If $V_0 = 12$ V, and the resistances of bulbs A and B are $5\ \Omega$, what current flows if the switch in the middle is open? (b) What current flows if the switch is closed, and through which bulb? *Hint: be careful, because the total current into a junction equals the total output current.*

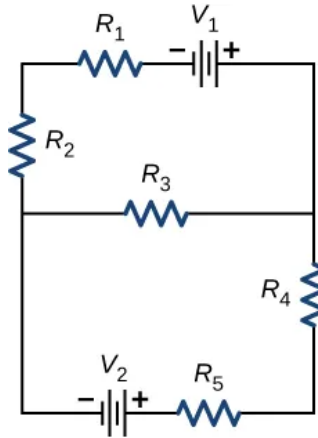


Figure 2: A circuit with two batteries and five resistors.

2. For the system in Fig. 1, what is the power consumption if the switch is (a) open, (b) closed?
3. Consider Fig. 2. Let $V_1 = 12$ V, $V_2 = 1.2$ V, and all resistors have a value of $2\ \text{k}\Omega$. (a) Use the loop rule to show that

$$V_1 - I_3 R_3 - I_1 R_2 - I_1 R_1 = 0 \quad (1)$$

- (b) Use a loop involving V_2 to create a similar equation as found in part (a) for the lower loop.

4. Solve for all currents through all resistors in Fig. 2.